

QUESTION PAPER

GitHub + Two-Stage CI Pipeline using Python

Number of Questions	3
Pipeline Requirement	Exactly 2 stages
Stage 1	Checkout Code
Stage 2	Build
Programming Language	Python

General Instructions

- For all three questions, create a Python project, push the code to GitHub, and configure a CI pipeline.
- The pipeline must contain exactly two stages: Stage 1 – Checkout Code and Stage 2 – Build.
- Use a Jenkins Declarative Pipeline (Jenkinsfile) unless your instructor specifies another CI tool.
- The Build stage should demonstrate execution of the Python project, such as installing dependencies and/or running the Python program.
- Submit the GitHub repository URL and the Jenkinsfile/pipeline configuration.

Question 1 – Basic Python Application

Create a simple Python application that accepts two numbers and displays their sum. Push the Python source code to a new GitHub repository. Configure a two-stage CI pipeline with: (1) Checkout Code – clone/checkout the GitHub repository. (2) Build – execute the Python program and verify that it runs successfully.

Requirement: Expected outcome: The pipeline should successfully check out the code and execute the Python application in the Build stage.

Question 2 – Python Calculator

Develop a Python calculator program that performs addition, subtraction, multiplication, and division. Store the source code in GitHub and create a Jenkinsfile for a two-stage pipeline. Stage 1 must check out the code from GitHub. Stage 2 must build/run the Python project and display the calculator output in the Jenkins console.

Requirement: Expected outcome: The pipeline must contain only the two required stages and complete successfully.

Question 3 – Python Student Result Program

Create a Python program that accepts marks for three subjects, calculates the total and average, and prints PASS or FAIL based on a suitable pass criterion. Push the project to GitHub. Create a two-stage

Jenkins pipeline where Stage 1 checks out the GitHub code and Stage 2 builds/runs the Python application. Ensure the Jenkins console shows the calculated result.

Requirement: Expected outcome: Demonstrate successful GitHub checkout followed by successful Python build/execution.

Pipeline Structure Expected for Every Question

Stage 1: Checkout Code → Checkout the Python project from GitHub

Stage 2: Build → Install required dependencies (if any) and run/verify the Python application